JAY AGARWAL

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EDUCATION

- **Birla Institute of Technology and Science, Pilani** *Nov 2022 - Aug 2027*
  M.Sc. Physics and B.E. Computer Science | *CGPA 8.1/10* Pilani, India
  – Coursework: Deep Learning, Introduction to Large Language Models, Group Theory & Applications, Probability & Statistics, Statistical Mechanics, Theory of Relativity, Quantum Mechanics I & II, Nuclear & Particle Physics, Computational Physics, Data Structures & Algorithms, Design & Analysis of Algorithms

PUBLICATIONS

- **What Do Lorentz-Equivariant Jet Taggers Learn?** J. Agarwal, S. Khare, D. Kumar. AI4Physics Workshop, ICML 2026. arXiv:2606.21790

- **What Transfers from a Lorentz-Equivariant Jet Tagger?** Under review, Representations for Physical Sciences (R/PS) Workshop, NeurIPS 2026.

RESEARCH EXPERIENCE

- **Visiting Researcher - KEK Theory Center** *Aug 2026 - Nov 2026*
  Host: Prof. Mihoko Nojiri Tsukuba, Japan
  – Funded research visit as part of my thesis on machine learning for high-energy physics, applying symbolic regression and interpretability methods to high-energy physics problems.

- **Undergraduate Researcher - PhantomGrad Lab** *Oct 2025 - Present*
  Advisor: Prof. Dhruv Kumar Pilani, India
  – Applied mechanistic interpretability to the Lorentz-Equivariant Geometric Algebra Transformer (L-GATr); implemented multivector grade ablations, boost equivariance tests and linear probes on scalar and multivector channels, with multi-seed runs and bootstrap confidence intervals.
  – Implementing knowledge distillation to compress equivariant L-GATr into a non-equivariant baseline, evaluating what structure transfers and at what deployment cost to accuracy.

- **Research Collaborator - TAMBO Collaboration** *Apr 2026 - Present*
  Advisor: Prof. Tommaso Dorigo INFN Padova (Remote)
  – Optimizing detector layout for the TAMBO neutrino observatory using differentiable programming; performed landscape analysis over the design space and benchmarked surrogate models against ground truth.

- **Research Intern - Solid State Physics Lab, DRDO** *May 2024 - Jul 2024*
  Supervisor: Dr. Poornendu Chaturvedi | *Quantum Sensing Group* Delhi, India
  – Simulated coherent population trapping signals for atomic frequency standards and built a Python tool for absorption and transmittance spectra, adopted into the group's research workflow.

INDUSTRY EXPERIENCE

- **Quantitative Analyst Intern, Wells Fargo** *May 2026 - Jul 2026*
  Causal Inference Team Bengaluru, India
  – Automated the marketing model development pipeline, replacing manual setup, review and diagnostics with a standardized reproducible workflow with room for custom extensions.

SELECTED PROJECTS

- **Beating Hallucinations via Uncertainty-Conditioned Attention Gating** *Jan 2026 - May 2026*
  Course Project - Introduction to LLMs
  – Inference-time framework combining semantic entropy probes on hidden states with lookback-ratio attention grounding to detect hallucinations without retrieval or multiple decoding passes.
  – Implemented head-adaptive gating that suppresses ungrounded attention heads conditioned on predicted risk; evaluated on XSum, SQuAD, TriviaQA and Natural Questions.

- **Detecting Quantitative Spatial Inconsistencies in Vision-Language Outputs** *Feb 2026 - April 2026*
  Course Project - Deep Learning
  – Method for surfacing quantitative spatial inconsistencies in vision-language model outputs by reasoning over residual representations as a graph.

SKILLS

- **Languages & Tools:** Python, C, C++, PyTorch, JAX, NumPy, SciPy, Scikit-learn, Pandas, PyG, Git, Linux